

Empowering ABC Foodmart: Real-Time Insights for a Growing Future with Data

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Introduction

ABC Foodmart, a rapidly growing neighborhood grocery chain founded by CEO Mark Doe, has achieved early success through the opening of a second store within three months of launching operations on March 1st, 2024. The company's growth potential attracted external investors from a PE firm with access to real estate, leading to a significant capital infusion intended to support the opening of three additional locations in Brooklyn within 6 months.

However, this period of growth also raised some critical operational inefficiencies. Mark, who had focused strongly on day-to-day operations rather than strategy faced pressure from the new investment board to track financial performance of each location. ABC Foodmart's existing data infrastructure consisted of multiple spreadsheets used to track sales, purchasing, inventory and staffing. These manual methods limited the company's ability to perform integrated business analysis, resulting in inconsistent reporting, delays in decision-making and vulnerability to human error. Furthermore, the historical operating expense data was unavailable due to previous outsourcing of accounting functions without a reliable retention strategy to retrieve data.

ABC Foodmart's new management recognized that the company lacked the data integrity, visibility and analytical capabilities to manage profitability and required a consulting team to reinvent their database management system.

(Note: The dataset utilized for this project was simulated for educational purposes, but all analyses and deliverables were developed under the assumption that the data was client provided. A detailed explanation of the simulated data is available at: [Group5 Data Generation](#))

Business Needs Assessment

Based on initial discussions with stakeholders and evaluation of the existing environment, new management's focus on financial return of the business created several business needs were identified:

- Profitability monitoring: Ability to perform frequent, ad-hoc profitability checks at store and product levels
- Product performance: Analysis of popular and high-margin products to inform stocking and marketing strategies

- Returns management: Systematic tracking of product returns to assess financial impact and identify quality issues
- Supplier evaluation: Quantitative metrics to assess supplier reliability
- Seasonal trend analysis: Understanding sales seasonality to optimize inventory and staffing levels
- Expense recording and monitoring: Structured tracking of operational expenses to support comprehensive financial analysis

Overall, all needs required a scalable data platform capable of automating calculations, delivering real-time insights and supporting operational decision-making.

Proposed Solution and Implementation Plan

To address the identified needs, the team proposed a multi-phased approach centered on the deployment of a relational database system using PostgreSQL. The core elements of the solution included:

- Consolidating multiple spreadsheets into a centralized database that supports standardized, integrated data capture
- Structuring the database to meet Third Normal Form (3NF) requirements, ensuring minimized data redundancy and optimized query performance
- Building an Extract, Transform, Load (ETL) pipeline using Python to migrate current data into the new database system
- Developing pre-existing views to deliver quick access to frequently requested management reports such as profitability, COGS breakdown and labor costs
- Implementing triggers to automate operational updates, including sales recalculations, inventory adjustments and delivery status updates
- Establishing role-based access controls, providing analysts, managers and administrators with appropriate levels of data
- Deploying dashboards for management (using Metabase) that offer user-friendly, real time reporting capabilities

This solution was intended to transform ABC Foodmart with data-driven decision-making capabilities, operational transparency and a scalable technological foundation that aligned with the current and future expansion goals.

Team Contract and Timeline

Project execution was organized around clearly defined functional roles to ensure that all technical and business requirements are covered:

- Database Architect: Design the logical and physical database schema, enforce normalization standards, ensure referential integrity across the system
- ETL Developer: Lead the development of Python-based ETL scripts to execute data extraction, transformation and loading

- Data Analyst: Create and optimize complex SQL queries to extract insights from the database, addressing key business questions with actionable insights
- BI & Reporting Lead: Develop interactive dashboards using Metabase ensuring that visualizations are intuitive and aligned with stakeholder information needs

(A detailed breakdown of individual responsibilities is provided in the Team Contract, Appendix A)

Source Data Overview

The initial phase of the engagement involved acquiring and assessing the historical operational data provided by ABC Foodmart. The data was contained within a series of Excel spreadsheets, reflecting the company's manual tracking methods. There was a total of 4 spreadsheets with the following data: records to employee and staffing, product and supplier details, purchase orders, sales transactions and customers.

(A detailed schema of the client data is available in Appendix B)

Data Characteristics and Challenges

Analysis of the provided data revealed characteristics of non-integrated data sources:

- Format: Data existed in flat-file structures with limited use of relational linking
- Inconsistencies: Variations in naming conventions (i.e., productname vs. product_name), status descriptions and date formats
- Data Quality: Issues including missing values, unstructured but critical information within general notes fields (such as product returns)
- Integration: Linking related information required manual cross-referencing between files (i.e., purchase orders to supplier contracts)

The provided datasets formed the basis for designing the database schema and developing the ETL pipeline. While historical operating expense details were limited, the available data of sales/procurement/products/suppliers/employees provided sufficient data to build a foundational database supporting core operational analysis and future data capture requirements.

Relational Schema Overview

The schema was formed by identifying functional dependencies within the client-provided data, and decomposing the larger, denormalized structures into smaller, well-defined tables that represent a single entity or relationship. The finalized PostgreSQL database schema included 16 interconnected tables, designed to logically represent the operations of ABC Foodmart. The tables were divided into operational clusters:

- Foundation Tables: *locations, suppliers, manufacturers*. These tables have no dependencies and are constantly referenced across other relations

- Product & Supply Chain: *products, purchase_orders, purchase_order_details, deliveries, inventory*. These tables track the process of ordering goods, tracking products and inventory in each location
- Sales & Promotions: *sales_orders, sales_orders_details, returns, promotions, customers*. These tables capture customer sale transactions, specific items sold, returns and information on promotional events
- Operations: *employees, staffing, expenses*. These tables track employees, their work schedules and operational costs (excluding COGS)

(Complete logical structure and relationships of the database schema are visualized in the Entity-Relationship Diagram (ERD) provided in Appendix C)

Design Enhancements

In addition, several additional elements were incorporated to optimize system performance and user experience:

- Creation of Views: Dedicated views were developed for net profit per location, monthly gross margin trends, and cogs by category. Views provide pre-aggregated, user-friendly outputs that can be recalled without rewriting a complex query.
- Trigger-Based Automation: Triggers were implemented to automate some key processes, including real-time delivery status updates, inventory level adjustments upon sale or delivery, automatic total sales recalculations and inventory expiration date setting.
- Expense Data Structure: Although the client did not have previous expenses, an expenses table was introduced to being recording and analyzing operational costs.

ETL Process Implementation

The database schema was implemented in PostgreSQL using standard SQL CREATE TABLE statements. Key implementation choices were made to verify data integrity, performance and an appropriate representation:

- Data types: Careful consideration was given to each data type selection. NUMERIC was used for monetary values (i.e., *unit_price, unit_cost, amount*) to ensure precise decimal storage. DATE was used for date-only values (i.e., *order_date, hire_date*) while TIME was used to track employee shifts. VARCHAR was used for text fields with appropriate length constraints (i.e., VARCHAR(100) for names) to balance storage and flexibility. INTEGER was used for unit counts and BOOLEAN to flag loyalty membership status.
- Primary Keys: The SERIAL was chosen for most primary keys (i.e., *location_id, employee_id, product_id*) This automatically creates an incrementing integer

sequence that guarantees unique identification for each record while simplifying the data ingestion process, compared to managing manual keys.

- Foreign Keys: Relationships between these tables are explicitly defined using foreign keys to ensure referential integrity. For example, each record in *sales_orders_details* must link to a valid record in *sales_records* and *products* to prevent data loss.
- Additional constraints:
 - UNIQUE constraints were applied to business keys (i.e., *locations.location_name*) to prevent duplicate entries
 - NOT NULL constraints were used for mandatory fields (i.e., *products.product_name*, *sales_orders_details.quantity*) to ensure essential data was always captured
 - CHECK constraints were implemented to enforce specific rules on columns (i.e., *purchase_orders.status* IN ('Ordered', 'Received', 'Cancelled')) to further enhance data validity beyond simple type checks

These design choices collectively create a normalized, reliable database structure optimized for data integrity and efficient querying, directly addressing the client's limitations on using the previous spreadsheet-based system.

(Full schema is available at [Group5 SQL Table Insert.sql](#))

ETL Process

A custom ETL pipeline was developed to facilitate the migration of the client's structured and semi-structured data into the relational database:

- Extraction: Data from four Excel workbooks was parsed using the Python Pandas
- Transformation: Attribute names were mapped, missing values were handled, and data types were standardized to align with database schema requirements
- Load: Data was sequentially inserted into the database, respecting foreign key dependencies to maintain referential integrity

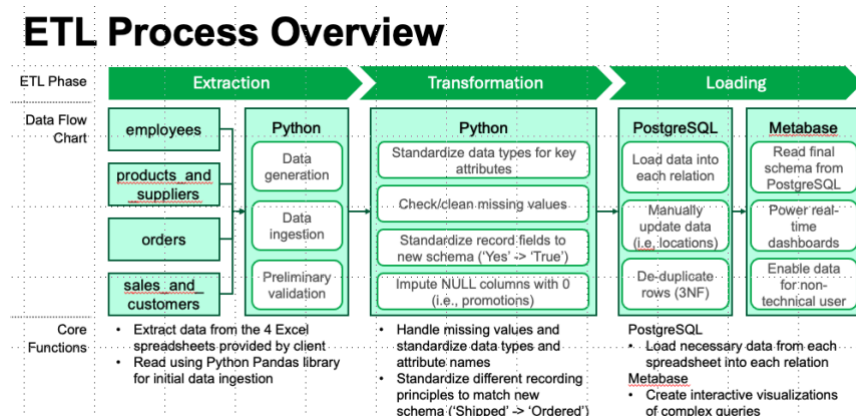


Figure 1: ETL Process Flow Diagram

(Python scripts detailing the ETL process are available at [Group5 ETL Pipeline](#))

Analytical Procedures

To address ABC Foodmart's most urgent business questions, the team conducted a needs-based prioritization of insights in close alignment with the client's specific goals: improving profitability, reducing inefficiencies, improving operations and increasing customer retention. Therefore, 4 critical analytical pillars were identified – Financial performance, Sales & Seasonal Trends, Supplier Reliability, and Customer Loyalty – and developed 10 procedures to unlock value across these areas.

1. Financial Performance: support tracking of profitability and costs
 - Net Profit per Location: Provides granular profitability analysis across store locations
 - Monthly Gross Margin per Location: Tracks gross margins over time to detect emerging financial trends
 - COGS Breakdown by Product Category: Highlights categories with the highest cost burdens
 - Estimated Labor Costs as a % of Net Revenue: Provides an efficiency metric by comparing estimated salary costs to net revenue per location
2. Sales & Seasonal Trends: uncover product and time-based revenue drivers
 - Top Categories by Net Revenue: Identifies high-performing product categories in the last 12 months based on net revenue after returns and discounts
 - Sales Volume Seasonality for Key Items: Analyzes season sales quantities for specific products to optimize inventory and staffing decisions
3. Supplier Reliability: enabling data-driven vendor management
 - Return Rate of Supplier products: Quantifies the financial and operational impacts of returned goods to quantify suppliers
 - Supplier Rating: Ranks suppliers based on historical delivery timeliness to support vendor management
4. Customer Loyalty: strengthening long-term growth through retention
 - Customer Repurchase Rate: Measures customer retention by calculating percentage of customers who made more than one purchase within a timeframe
 - Sales Contribution for Loyalty Members: Determines the sales attributed to registered loyalty members versus guest checkouts, to assess the loyalty program

(Full SQL code for each procedure is available at: [Group5 Analytical Procedures](#))

Business Intelligence and Dashboards

To support executive decision-making, an interactive Metabase dashboard was developed that visualizes profitability metrics for each location. The dashboards present net revenue, costs of goods sold, labor expenses and net profit in real-time, allowing leadership to track each store's financial performance without requiring direct database access. By transforming complex financial data into intuitive visual summaries, the Metabase dashboard will empower management to identify operational strengths and weaknesses across locations, quickly respond to emerging trends and support strategic expansion decisions.

(Dashboard structure can be seen in Appendix D)

User Access Model and Security Strategy

A role-based access model was implemented to ensure appropriate data governance to balance operational needs with necessary security and privacy controls:

- Analysts: Grant read-only access to dedicated Views to enable querying without risking transactional data integrity
- Managers & Executives: Provide dashboard access only, facilitating quick performance assessments without direct interaction with the database
- HR/Administrators: Grant special access to sensitive information such as employee salaries and staffing schedules

To further enhance security, the server would have a database user authentication that requires each user role to be associated with login credentials protected by strong password policies. In addition, administrative actions, logins and data access would be logged as a backup for audits or version controls of the data.

Conclusions

With the implementation of a new database system, ABC Foodmart now can monitor critical financial, operational, supplier and customer metrics that were not being tracked. This has enabled a level of operational insight and data-driven decision-making essential for sustainable growth.

Financial Performance. During its first years of operations, ABC Foodmart is still operating at a net loss of approximately 10-15% of total revenue across locations. Even with the CEO working full time at the East store, he primary contributor to current losses is employee salary costs. However, the financial performance is still notably outperforming the industry average for new retail operations that often experience 20-30% losses when scaling. Given the pace of revenue growth, ABC Foodmart is positioned to

achieve its breakeven point relatively soon, assuming it can sustain sales momentum and further optimize operations.

Sales Trends and Seasonality. Sales analysis revealed that Pantry, Frozen and Beverages were the top-performing categories in terms of net revenue. Seasonal product trends aligned predictably with consumer behavior. For example, Vanilla Ice Cream sales peaked during the summer months, while Diet Cola maintained strong, steady sales throughout the year. These insights provide valuable opportunities for optimizing purchase orders and reducing COGS costs, while optimizing staffing schedules for seasonal demand patterns.

Supplier Insights and Operational Improvements. Supplier analysis highlighted important operational improvements made possible by the new database system. Several suppliers consistently exhibited delayed deliveries, affecting supply chain reliability. The introduction of supplier rating metrics -based on return rates and delivery performance- represents a significant leap in operational efficiency, allowing ABC Foodmart to renegotiate with suppliers based on factual data and minimize revenue loss from lacking inventory.

Customer Loyalty and Growth Potential. The most compelling finding from the data was in customer behavior. Customer repurchase rates across locations exceeded ~60%, almost 2-3x (20-40%) than the typical industry standard. This exceptionally high repurchase rate, combined with a significant proportion of sales attributed to loyalty program members, indicates a highly engaged and satisfied customer base. It is likely that this strong customer loyalty has played a foundational role in attracting external investor interest and will continue to propel ABC Foodmart's growth.

Future Recommendations

Going forward, to extend the value of the existing database, ABC Foodmart should implement product-level promotion ROI tracking by linking promotions to sales performance metrics such as unit lift, margin change and baseline comparisons. This would enable precise evaluation of campaign effectiveness across categories for management. Additionally, enriching the supplier rating schema with fields for unit cost volatility and average lead time would account for pricing stability and fulfillment reliability. These key metrics can be used for renegotiating vendor contracts. Finally, by building a loyalty scoring framework based on customer behavior (i.e., frequency, basket size), ABC Foodmart can segment its members and deploy personalized, tiered loyalty to strategies to further strengthen retention and lifetime value.

Group 5 ABC Foodmart Team Contract

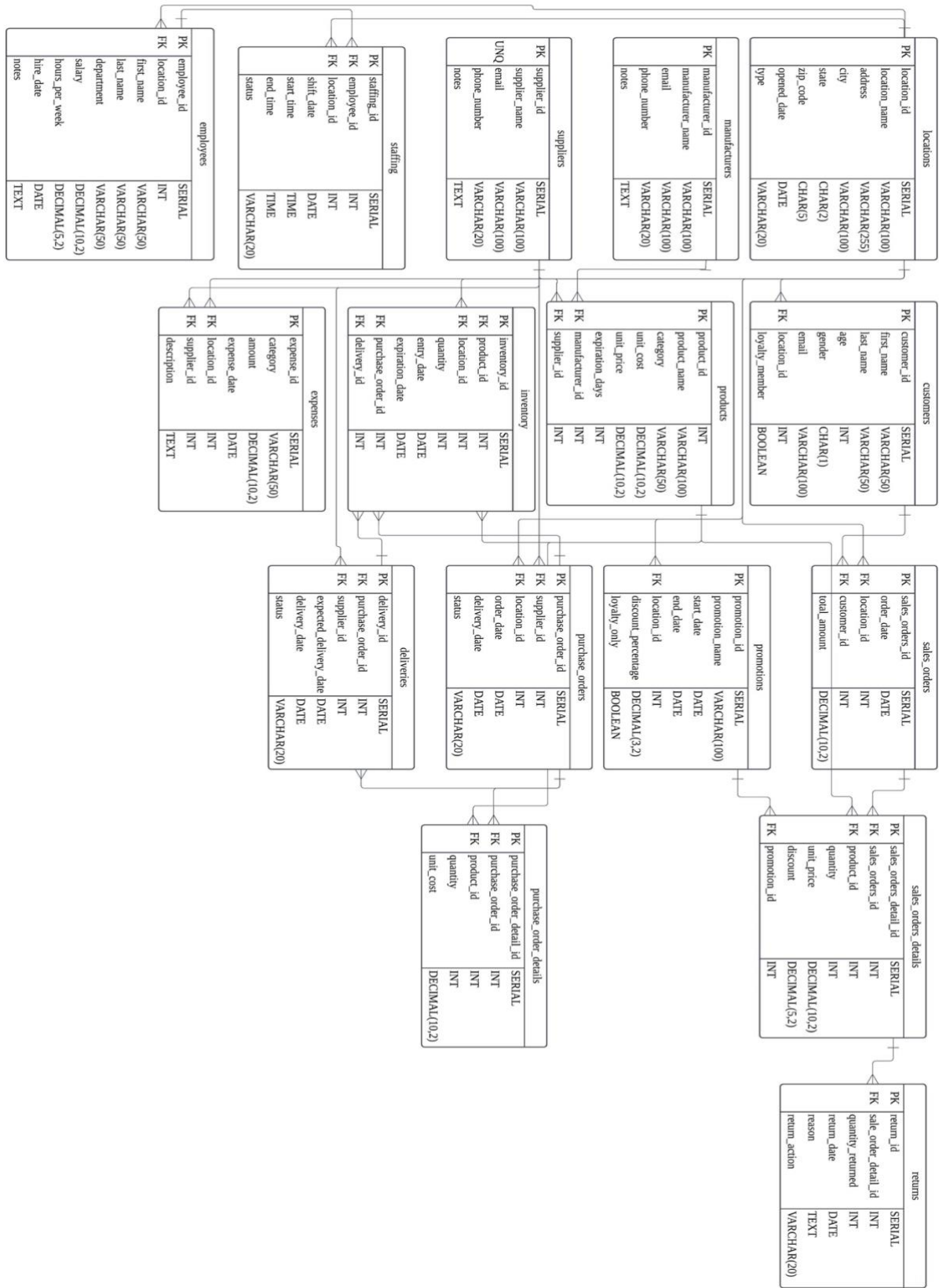
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Deliverables				
Task Bucket	Task	Responsible Member	Timeline	Status
1. Database Design & Schema	Develop a logical schema for 15 tables in 3NF	Jaejun Lead	Week 1	Complete
	Define primary/foreign keys and table relationships	Jaejun Lead	Week 1	Complete
	Finalize ER diagrams and normalization documentation	All team members	Week 1	Complete
	Conduct a peer review and refine the schema design	All team members	Week 1/2	Complete
2. Data Integration & Testing	Create and integrate sample data into the database	Min Sung Lead	Week 2	Complete
	Run initial tests for data accuracy and performance	Min Sung Lead	Week 2	Complete
	Troubleshoot and refine data integration process	All team members	Week 2	Complete
	Conduct testing to verify seamless data flows	All team members	Week 2/3	Complete
3. Dashboard & Reporting Design	Define key performance indicators (KPIs) and metrics for dashboards	Naniilkham Lead	Week 2/3	Complete
	Identify data visualization tools and plan integration	Naniilkham Lead	Week 2	Complete
	Develop prototype queries to extract unified data	Naniilkham Lead	Week 2/3	Complete
	Gather feedback on dashboard design and finalize	All team members	Week3	Complete
4. Documentation, Project Management & Final Review	Compile detailed report on design decisions	Sara Lead	Week 3	Complete
	Hold a final review meeting and submission preparation	Sara Lead	Week 3	Complete
	Prepare final presentation slides and supporting materials	All team members	Week 4	Complete

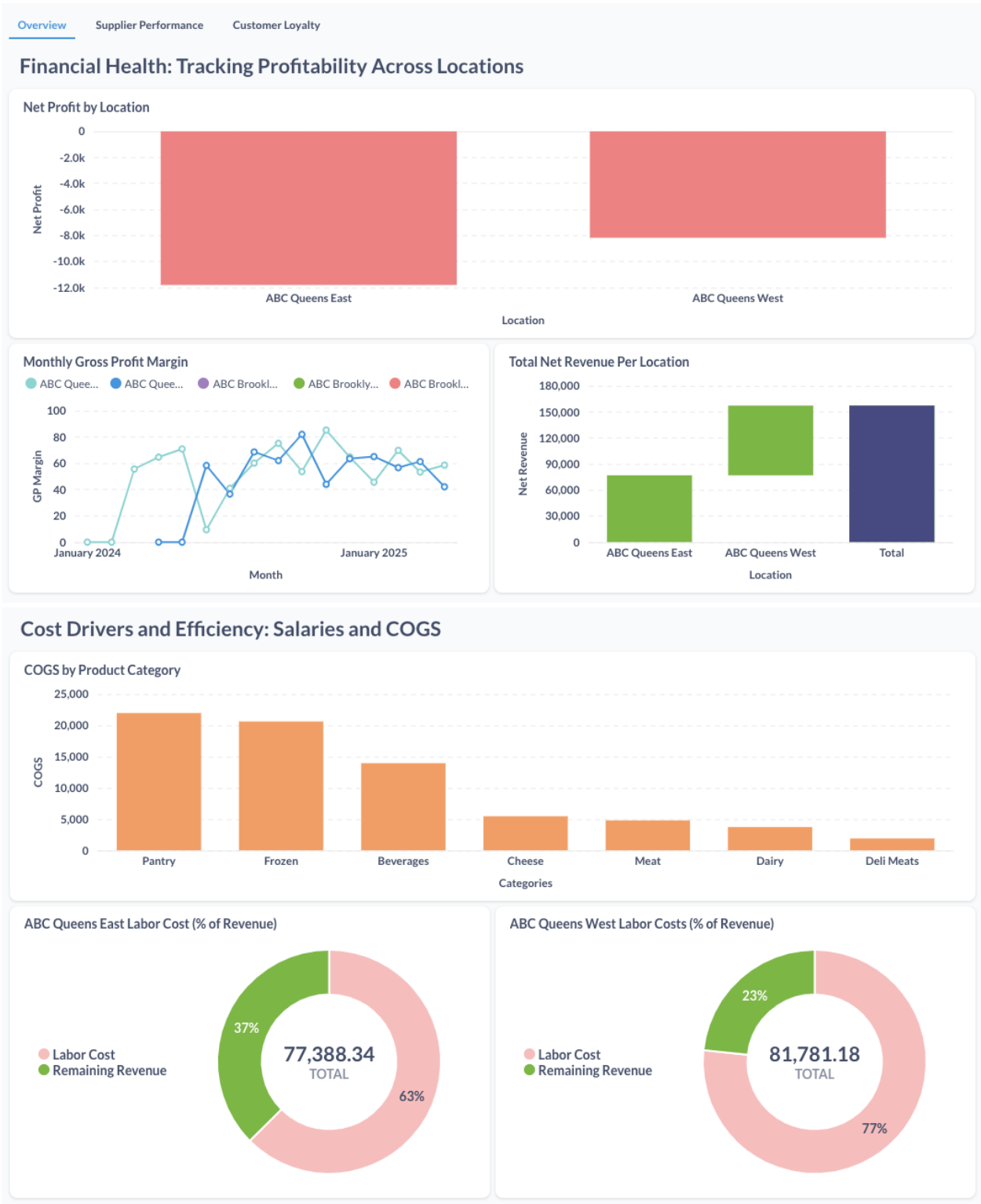
Appendix B – Client Data Schema

ABC_employees.xlsx	first_name	ABC_products_and_suppliers.xlsx	productname
	last_name		category
	department		unitcost
	salary		unitprice
	hours_per_week		expiration_days
	hire_date		manufacturer_name
	location_name		supplier_name
	shift_date		supplier_email
	start_time		supplier_phone
	end_time		notes
shift_status			
ABC_orders.xlsx	order_date	ABC_sales_and_customers.xlsx	sale_date
	expected_delivery_date		location_name
	delivery_date		customer_first_name
	supplier_name		customer_last_name
	location		age
	status		gender
	productname		email
	units		loyalty_member
	unit_cost		product_name
			quantity
	unit_price		
	discount_applied		
	promotion_name		
	notes		

Appendix C – Complete ERD Diagram



Appendix D – Metabase Dashboard

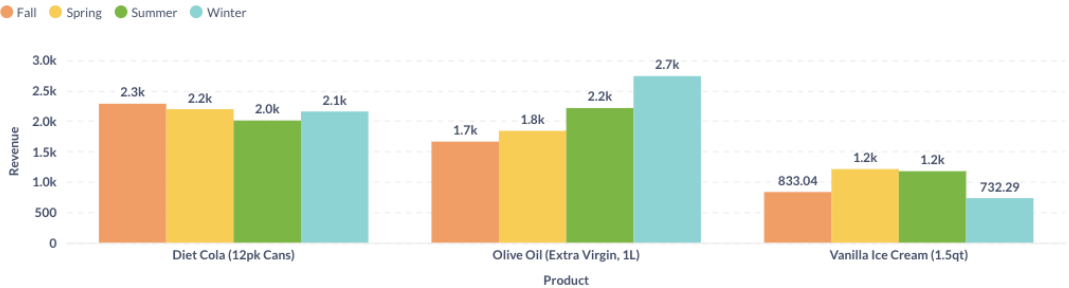


Revenue Dynamics: Share Distribution and Seasonality

Top Categories by Net Revenue



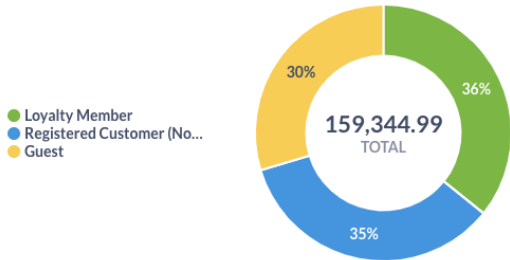
Seasonal Sales Volume (by Top Products)



Overview Supplier Performance Customer Loyalty

Customer Loyalty Insights: Driving Repeat Sales and Growth

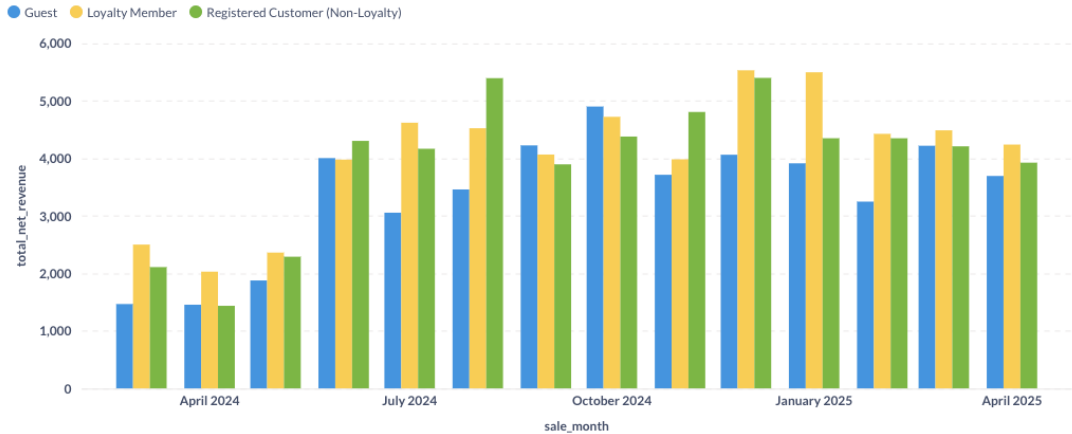
Net Revenue Distribution by Customer Loyalty



79.84%

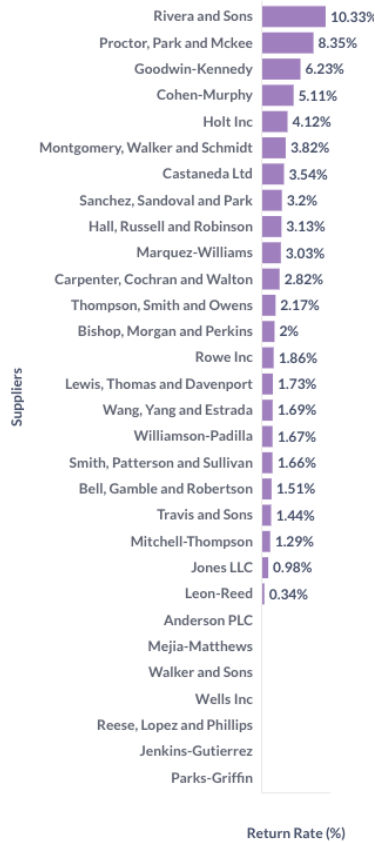
Customer Repurchase Rate

Loyalty vs Guest Sales Over Time

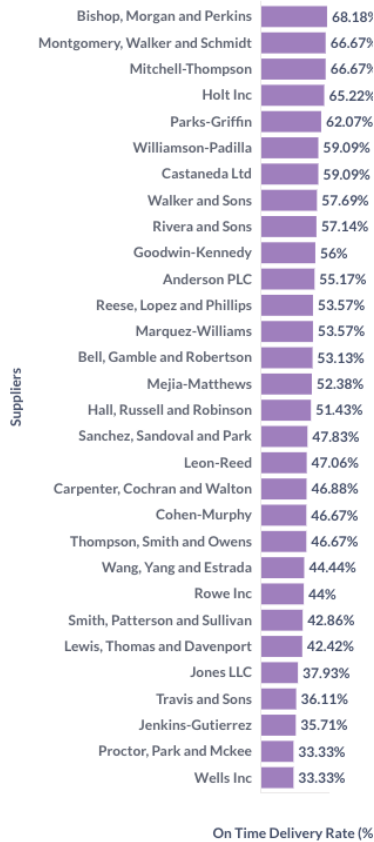


Vendor Scorecard: Returns, Deliveries and Overall Ratings

Return Rates (%) by Supplier



On Time Delivery Rate (%) by Supplier



Overall Supplier Rating

